K-NN numericals-

Q 1 - Apply k-means clustering (with k = 3) on the following dataset of 2D points. Use the Euclidean distance metric and initialize the centroids at
$C_1 = (1, 2)$, $C_2 = (3, 5)$ and $C_3 = (6, 4)$

Points	P1	P2	P3	P4	P5	P6
X	1	2	3	5	6	7
Y	2	4	6	5	4	3

Given

Points:
P1(1,2), P2(2,4), P3(3,6), P4(5,5), P5(6,4), P6(7,3)

Initial Centroids:

- C1 = (1,2)
- C2 = (3,5)
- C3 = (6,4)

k = 3

Step 1: Distance Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Iteration 1: Assign Points to Nearest Centroid

Point	C1 (1,2)	C2 (3,5)	C3 (6,4)	Cluster
P1(1,2)	0	3.6	5.38	C1
P2(2,4)	2.23	1.41	4	C2
P3(3,6)	4.47	1	3.6	C2
P4(5,5)	5	2	1.41	C3
P5(6,4)	5.38	3.16	0	C3
P6(7,3)	6.08	4.47	1.41	C3

Clusters after Iteration 1

- **C1:** P1
- **C2:** P2, P3
- **C3:** P4, P5, P6

◆ Step 2: Recompute Centroids

C1:

(1,2)

C2:

$$\left(\frac{2+3}{2}, \frac{4+6}{2}\right) = (2.5, 5)$$

C3:

$$\left(\frac{5+6+7}{3}, \frac{5+4+3}{3}\right) = (6, 4)$$

➢ Iteration 2: Reassign Points

Point	C1 (1,2)	C2 (2.5,5)	C3 (6,4)	Cluster
P1	0	3.35	5.38	C1
P2	2.23	1.12	4	C2
P3	4.47	1.12	3.6	C2
P4	5	2.5	1.41	C3
P5	5.38	3.64	0	C3
P6	6.08	4.92	1.41	C3

Clusters after Iteration 2

- **C1:** P1
- **C2:** P2, P3
- **C3:** P4, P5, P6

Clusters remain same → Converged

Final Clusters are

- **Cluster C1:** {P1} → Centroid (1,2)
- **Cluster C2:** {P2, P3} → Centroid (2.5, 5)
- **Cluster C3:** {P4, P5, P6} → Centroid (6,4)

Final Answer-

After applying K-Means clustering with k=3:

- **C1 = {P1}**
- **C2 = {P2, P3}**
- **C3 = {P4, P5, P6}**

Centroids: C1 = (1,2) , C2 = (2.5,5) , C3 = (6,4)

Q2 - Apply k-means clustering (k = 3) using Euclidean distance.

Initial centroids:

$C_1 = (2, 2)$, $C_2 = (5, 6)$, $C_3 = (8, 3)$

Dataset:

Points	P1	P2	P3	P4	P5	P6
X	2	3	4	6	7	8
Y	2	3	5	7	4	3

✓ Answer Key

- C1 → {P1, P2} → (2.5, 2.5)
- C2 → {P3, P4} → (5, 6)
- C3 → {P5, P6} → (7.5, 3.5)

Q3 Apply k-means clustering (k = 2)

Initial centroids:

$C_1 = (1,1)$, $C_2 = (6,6)$

Dataset:

Points	P1	P2	P3	P4
X	1	2	5	6
Y	1	2	5	6

✓ Answer Key

- C1 → {P1, P2} → (1.5, 1.5)
- C2 → {P3, P4} → (5.5, 5.5)

Q4 clustering (k = 3)

Initial centroids:

$C_1 = (1,3)$, $C_2 = (4,5)$, $C_3 = (7,2)$

Dataset:

Points	P1	P2	P3	P4	P5	P6
X	1	2	3	5	6	7
Y	3	4	5	5	3	2

✓ Answer Key

- C1 → {P1, P2} → (1.5, 3.5)

- C2 → {P3, P4} → (4, 5)
- C3 → {P5, P6} → (6.5, 2.5)

Q5 Apply k-means clustering (k = 2)

Initial centroids:

$C_1 = (2,8), C_2 = (7,3)$

Dataset:

Points	P1	P2	P3	P4	P5
X	2	3	4	6	7
Y	8	7	6	4	3

✓ Answer Key

- C1 → {P1, P2, P3} → (3, 7)
- C2 → {P4, P5} → (6.5, 3.5)

Q6 Apply k-means clustering (k = 3)

Initial centroids:

$C_1 = (1,2), C_2 = (5,5), C_3 = (9,2)$

Dataset:

Points	P1	P2	P3	P4	P5	P6
X	1	2	4	6	8	9
Y	2	3	5	5	3	2

Answer Key

- C1 → {P1, P2} → (1.5, 2.5)
- C2 → {P3, P4} → (5, 5)
- C3 → {P5, P6} → (8.5, 2.5)